6BUIS002 – Information Driven Entrepreneurship and Enterprise



Lecture 2 – E-business and e-commerce

Formulating information driven enterprises

Enterprises in the era of e-commerce and e-business

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Networking technologies at modern enterprises

E-business and e-commerce fundamentals

Further reading:

Dave Chaffey, Tanya Hemphill, and David Edmundson-Bird:
Digital Business and E-Commerce Management, *7th ed.*
Pearson Education, 2019.

Essential: Chapters 1-2

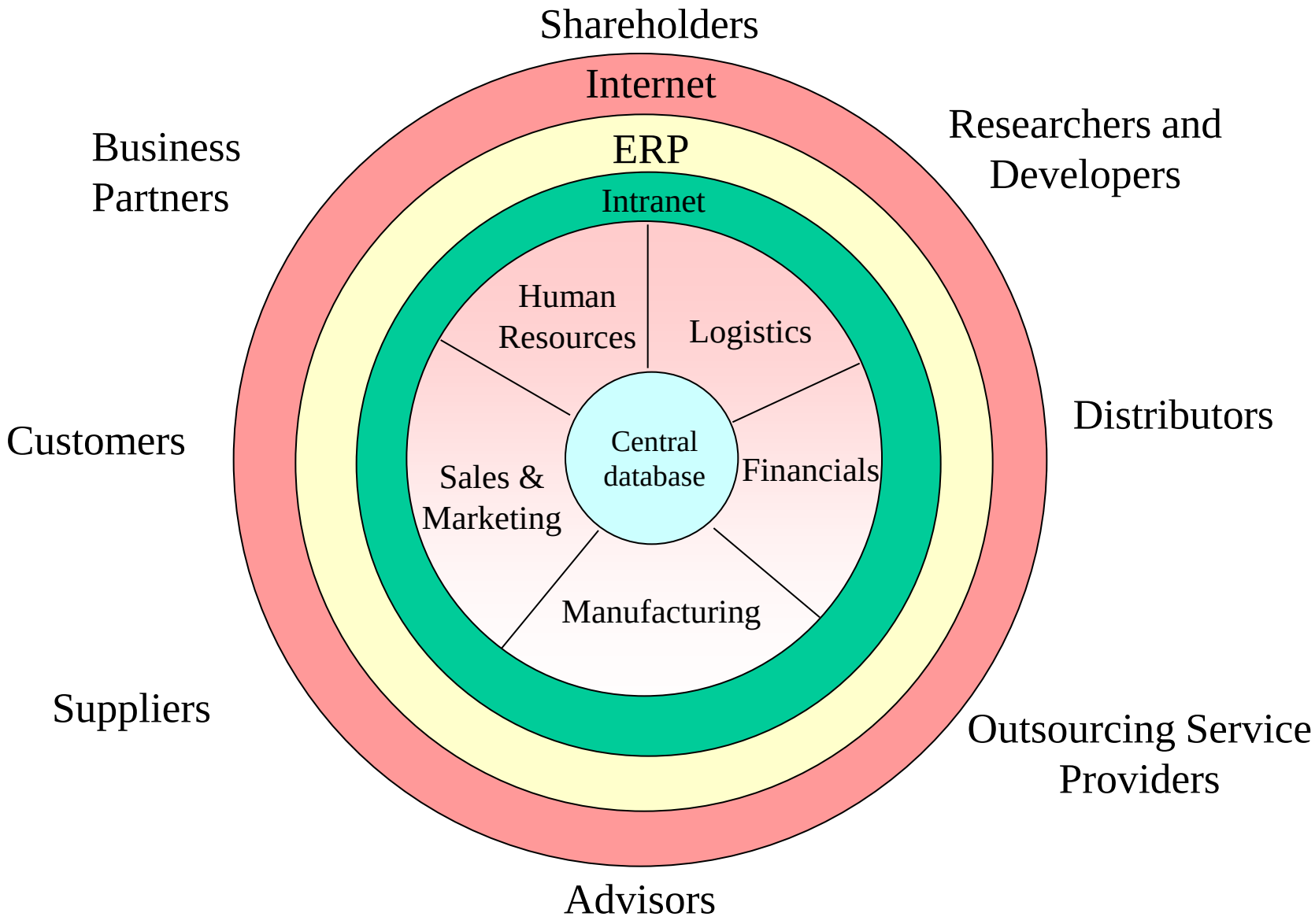
Recommended: Chapters 3-4

Book is available on-line via the Library Webpage.



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Reminder - Enterprise applications and networks



ERP: Enterprise Resource Planning

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Reminder - E-business and e-commerce defined

There are no universally accepted definitions – terms used interchangeably

E-commerce:

where business transactions (financial and non-financial) take place via telecommunications networks, especially the Internet

buying and selling of products, services, and information via computer networks including the Internet

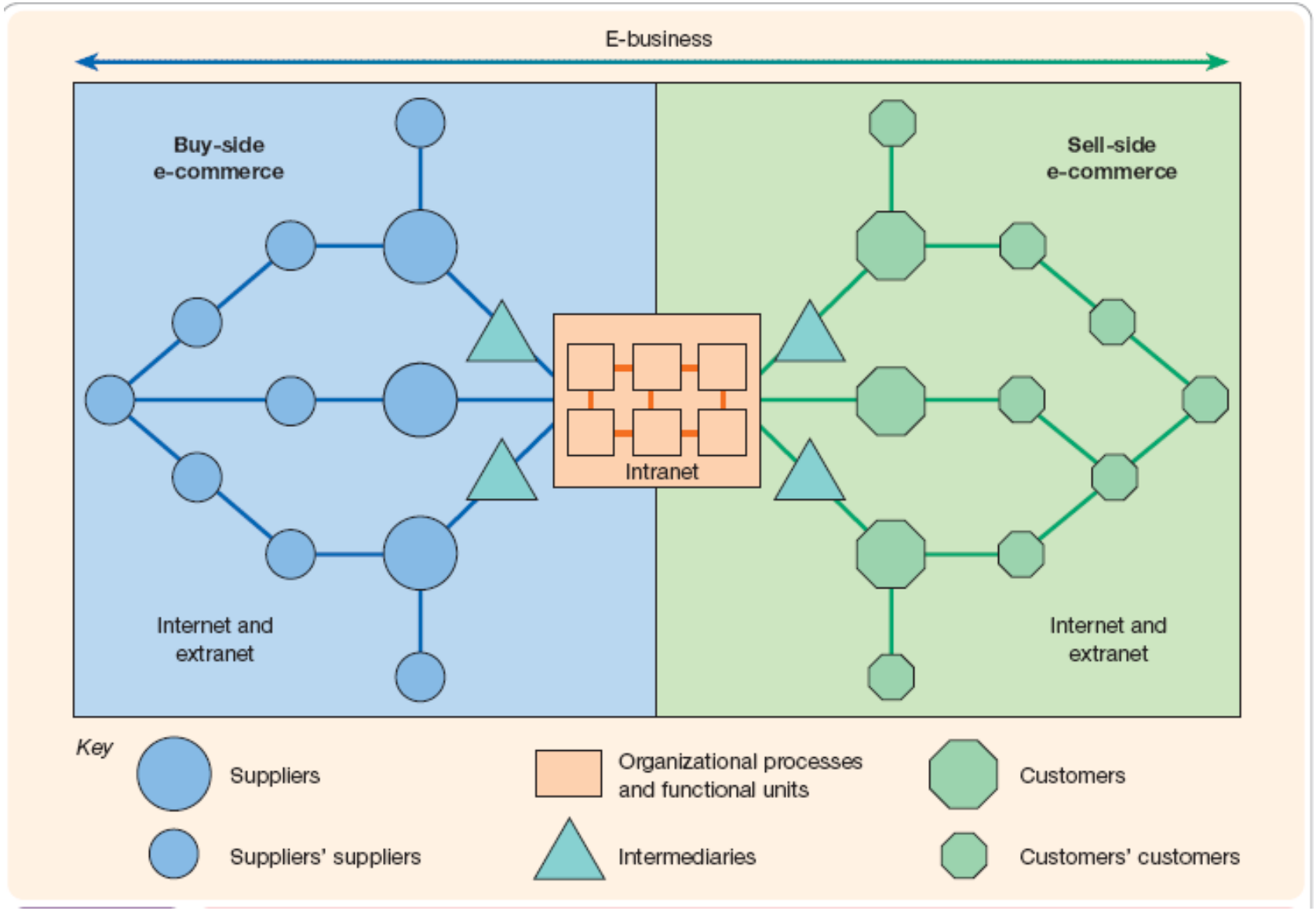
E-business:

The transformation of key business processes through the use of networking and Internet technologies

E-business describes the broadest definition of EC. It improves business performance by using electronic information technologies and open standards to connect suppliers and customers at all steps along the value chain.

E-business is more than EC transactions, it is the overall strategy of redefining old business models, with the aid of technology, to maximise customer value and profits.

Reminder - E-business and buy-side and sell-side e-commerce



Source: Chaffey 2009

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The 4 phases of utilising networking technology for business purposes

1. **Brochureware:**
   - the Internet is only used as a marketing tool through a static website
2. **Electronic commerce**
   - selling and buying goods on-line
3. **E-business**
   - intensive relationship between business partners through the network
4. **E-enterprise**
   - the whole value chain – from procurement of raw materials to customer relationship management – happens by combining traditional methods and networking technologies

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Classifications of E-Business

1. According to forms of operation



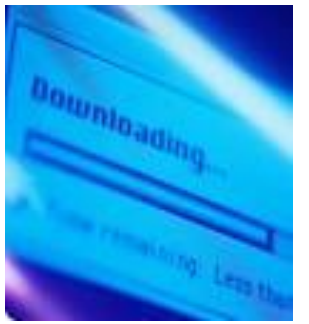
Dotcom business

Company operating only on-line

Dotcorp business (alias click-and-mortar, bricks-and-clicks)

Company operating both on-line and off-line

2. According to dimensions



Pure EC

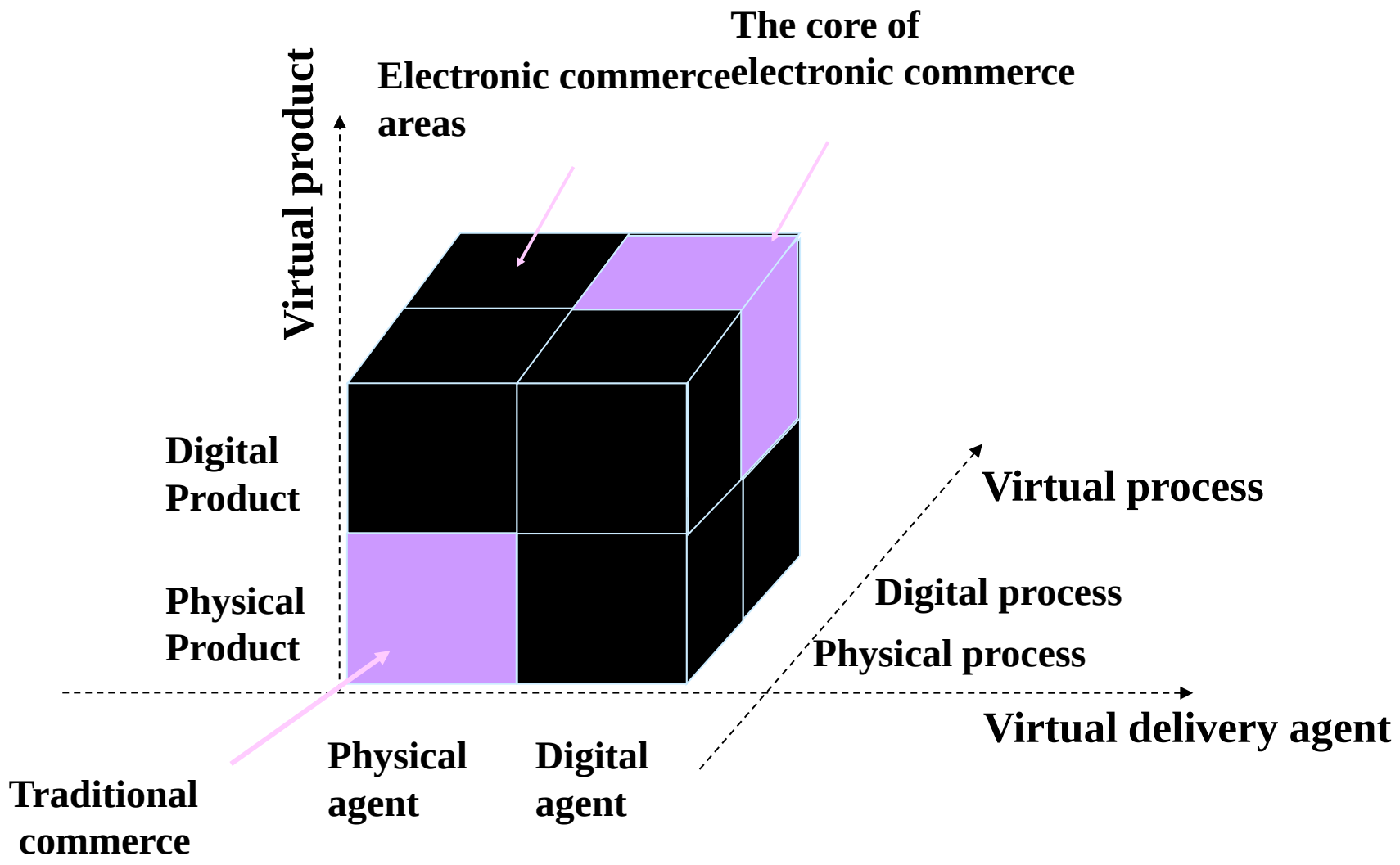
digital product, process & delivery

Partial EC

mixture of digital and physical dimensions

3. According to participants

Dimensions of E-Commerce



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Classification of E-Business according to participants

Simple version:

- B2B: Business-to-business
- B2C: Business-to-customer
- B2G: Business-to- government

Complex version:

To: / Initiated by:	Business (B)	Customer(C)	Government(G)
Business (B)	B2B	B2C	B2G
Customer(C)	C2B	C2C	C2G
Government(G)	G2B	G2C	G2G

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Classification of E-Business according to participants

B2B – Business to Business e-business:

- Commercial transactions between an organisation and other organisations

B2C – Business to Customer e-business:

- Commercial transactions between an organisation and consumers

B2G (B2A) – **Business to Government** (Business to Administration) **e-business:**

- Transaction between a business organisation and a governmental agency

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Classification of E-Business according the participants

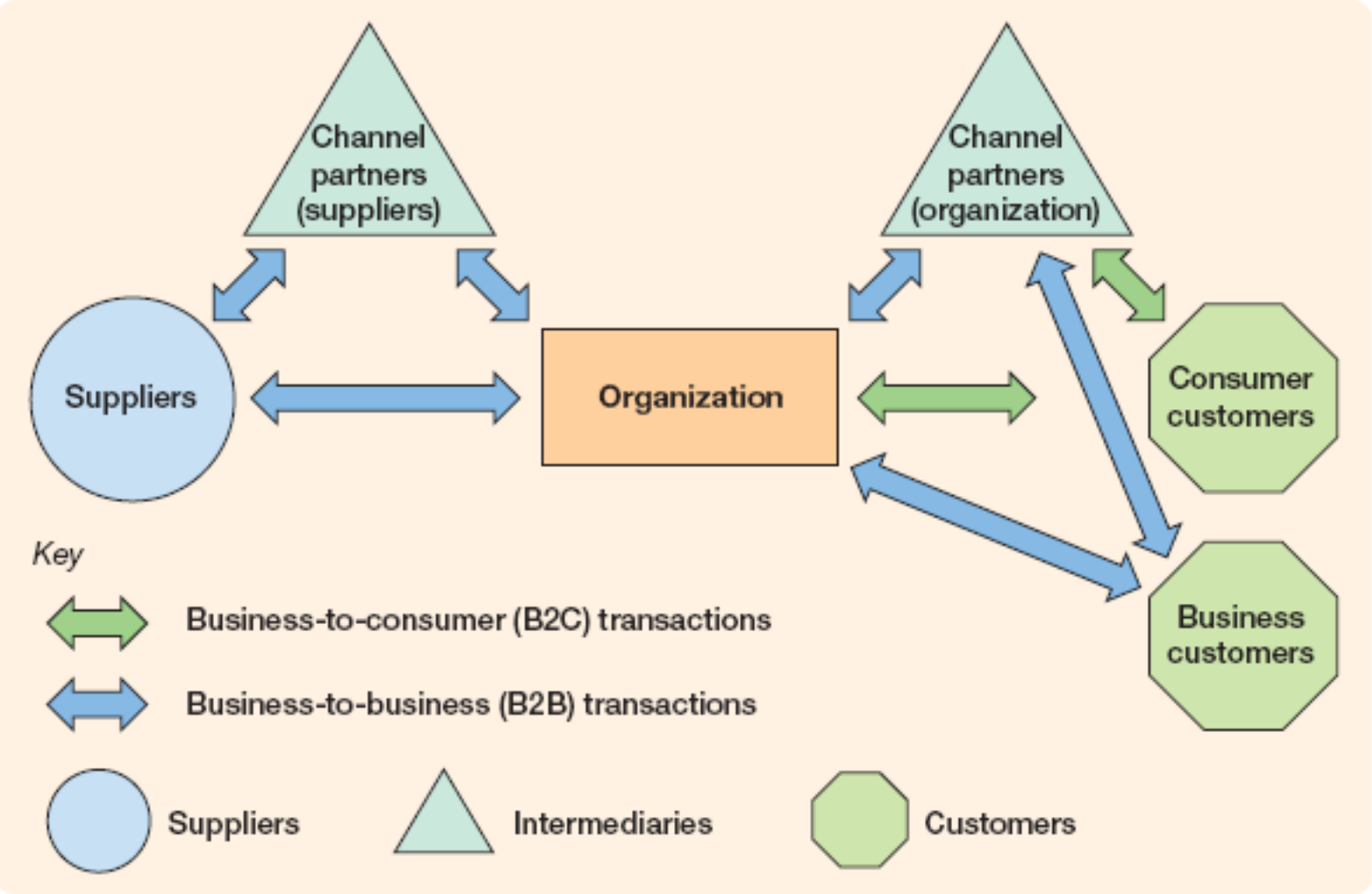
Examples

To: Consumer of content/service	From: Supplier of content/service: *Consumer or citizen*	From: Supplier of content/service: *Business (organization)*	From: Supplier of content/service: *Government*
Consumer or citizen	**Consumer-to-Consumer (C2C)** • eBay • Peer-to-Peer (Skype) • Blogs and communities • Product recommendations • Social networks: MySpace, Bebo	**Business-to-Consumer (B2C)** • Transactional: Amazon • Relationship-building: BP • Brand-building: Unilever • Media owner – News Corp • Comparison intermediary: Kelkoo, Pricerunner	**Government-to-Consumer (G2C)** • National government transactional: Tax – inland re venue • National government information • Local government services
Business (organization)	**Consumer-to-Business (C2B)** • Priceline • Consumer-feedback, communities or campaigns	**Business-to-Business (B2B)** • Transactional: Euroffice • Relationship-building: BP • Media Owned: Emap business publications • B2B marketplaces: EC21	**Government-to-Business (G2B)** • Government services and transactions: tax • Legal regulations
Government	**Consumer-to-Government (C2G)** • Feedback to government through pressure group or individual sites	**Business-to-Government (B2G)** • Feedback to government businesses and non-go vernmental organizations	**Government-to-Government (G2G)** • Inter-government services • Exchange of information

Source: Chaffey 2009

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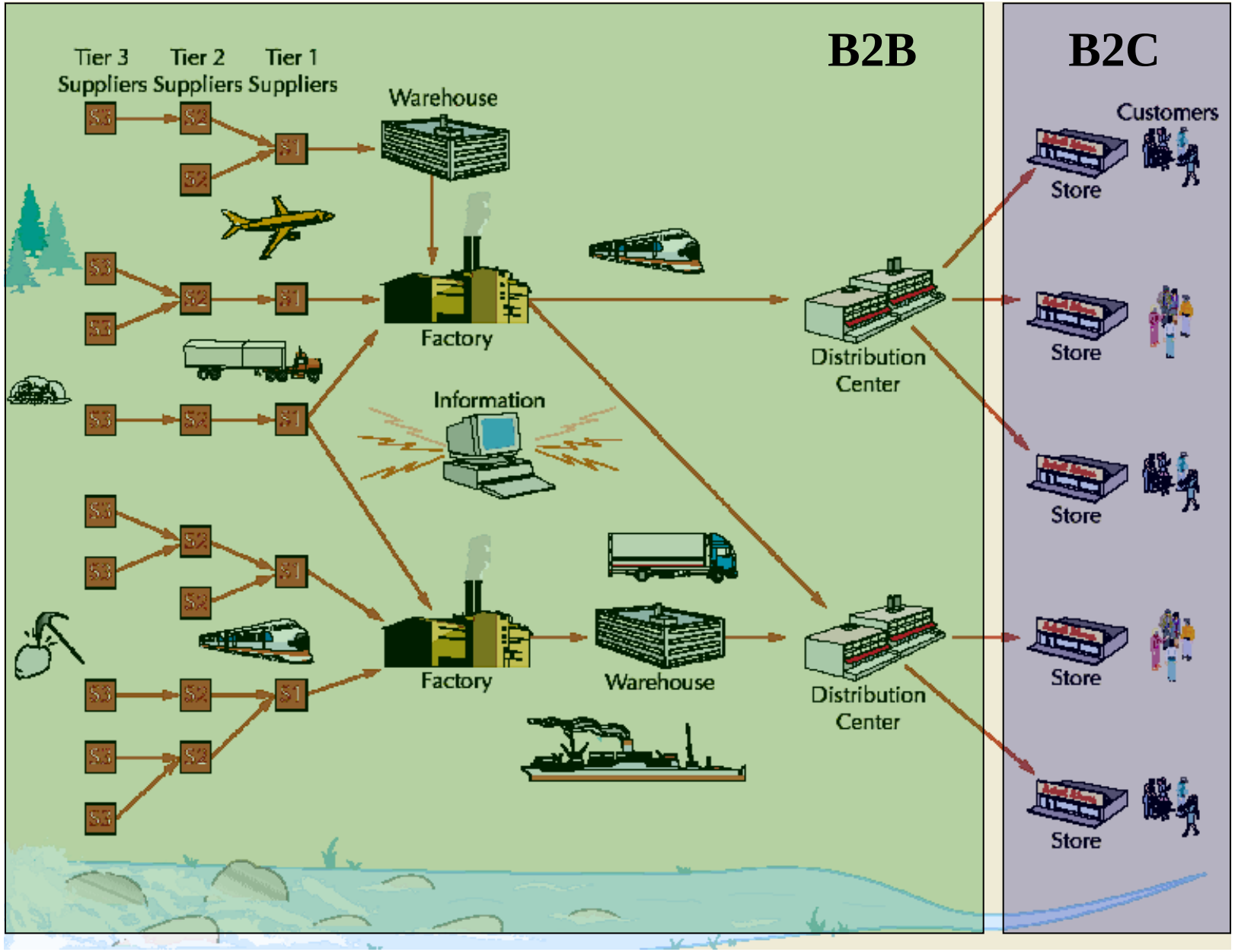
Business to Customer and Business to Business E-commerce



Source: Chaffey 2009

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Supply chain, B2C and B2B



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B2C e-Commerce

Definitions of B2C e-Commerce:

- Commercial transactions between an organisation and consumers
- E-commerce between companies and customers
- Retailing through the Web

Characteristics:

- Open 24 hours 7 days a week
- Global access
- Online catalogs, product information
- Online ordering

B2C (C2B, C2C) e-Commerce Models

Storefront model

- E-store (web store) E.g.: Apple.com
- Online shopping mall E.g.: Shopping.com

Portal Model E.g.: Yahoo.com

Free services E.g.: Freelotto.com

Auction Model
- E.g.: eBay.com

Dynamic-pricing models
- E.g.: Priceline.com

} Widely used in B2B as well

Storefront models

- E-store:
  - E.g.: www.hp.com, apple.com
  - Single merchant
- Online shopping mall:
  - E.g.: www.fashionmall.com
  - Products from several vendors
    - general
    - specialized





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Auction Model

Intermediates between seller and buyer.



E.g.: www.ebay.com

Types of auctions:

- ***English auction***
  - Ascending
  - Bidders increase the price against each other
  - Highest bid wins
- ***Dutch auction:***
  - Descending
  - Auctioneer starts high, slowly decreases price
  - First bid wins

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Auction Model

Types of auctions:

- **Reverse Auction Model**
  - Many sellers, one buyer
  - Buyer sets price
  - Sellers compete
  - E.g.:ewanted.com
  - Fastest: automated bidding system down to the seller's reserve price
  - Very common in B2B e-commerce too

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Portal Model

Horizontal portal

E.g.: Yahoo.com

- Integrates wide range of information and links to different services
- Typical content:
  - news, sports, weather information
  - search engine
  - links to e-stores and shopping malls
  - e-mail
  - discussion groups
  - blogs

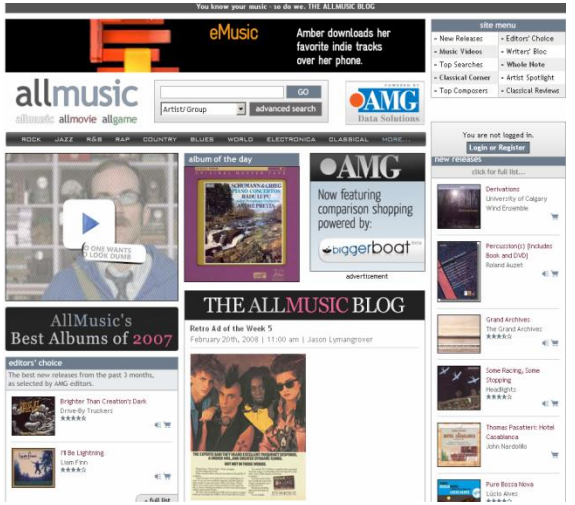


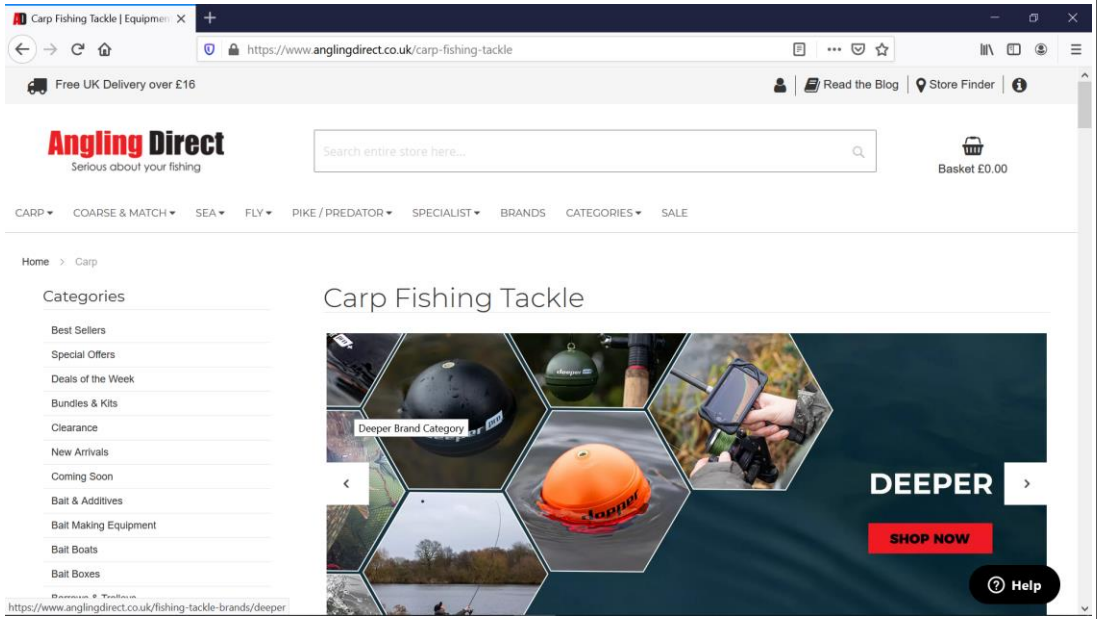
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Portal Model

Vertical portal

- E.g.: www.allmusic.com, www.anglingdirect.co.uk,
- Large amount of information and services pertaining to a single area of interest (profession, age, sports, hobbies etc)
- Vertical industry or market or specific group portal
- has growing importance because of the rapidly growing amount of online information
- Benefits:
  - Industry-specific information
  - Online directories, business opportunities
  - Online communities, forums, chat, blogs





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Problems of the portal model

High maintenance costs

- Content must be permanently updated

High innovative costs

- Always new attractions are needed

Financing problems

- Revenue is generated from advertisements / sponsors
- Too many ads result in loosing visitors

Social networking sites

- Similar and derived from portal model
- Solves some of the problems of portals

Maintenance costs

- Lower as users update content

Innovative costs

- Lower as user may also provide attractions

Financing

- Advertisements are tailored for specific interest

Dynamic Pricing Models

Name-your-price model (E.g.: www.priceline.com)

- Customer offers price for a product / service

Comparison-pricing model (E.g.: www.comparethemarket.com/)

- Finds the desired product / service for the customer at the lowest price within a wide range of e-stores

Demand-sensitive pricing model

- Collects demand from individual customers and buys product in bulk for them to get quantity discount

Bartering model (www.barteronly.com**)**

- Intermediates between customers wishing to exchange products

Rebates (E.g.: www.ebates.com)

- Offers cash refund and coupons after purchasing

B2B e-commerce

Definitions

- Commercial transactions between an organisation and other organisations
- Electronic business between organizations
- Trade / collaboration among business communities over the Internet

Significance of B2B e-commerce

– biggest volume of electronic trade by far is business-to-business (B2B) (when compared to B2C) - involving big firms and SMEs

B2B e-business models

- **Sell side:**
  - Using B2C sites for B2B purposes applying quantity discounts (both buy and sell)
  - Establishing corporate B2B distribution site and separate B2C site for direct selling
  - Forming a consortium to establish a B2B exchange site (e-marketplace)
  - Using existing B2B exchange sites (e-marketplaces) run by other companies, consortiums or intermediaries
- **Buy side:**
  - Using desktop purchasing systems
  - Creating corporate procurement portals

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Desktop purchasing systems

- Enable employees to purchase indirect goods and services from the Internet
  - Multivendor catalogues approved by management are stored on the intranet
  - Employee's order is automatically routed to boss for approval (workflow)
  - After approval order is sent to supplier
  - Supplier confirms and deliver
- Used at large firms (E.g.: Ford, Microsoft, UoW)

E-marketplaces

- a trading forum on the Internet in which multiple buyers and sellers exchange goods and services
- an e-marketplace is always primarily B2B





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Services Offered by Marketplaces

- Information exchange
  - Datasheets
  - Forums
  - Chats
  - Newsletters
- Trading
  - Enabling to create and run a storefront model
  - Auctioning
    - Forward or reverse auctions
    - Handling RFP (Request for Proposal), RFQ (Request for Quote)
    - Open or sealed bids
  - Bartering
- Collaboration
  - Allows companies to plan together

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Classification of E-marketplaces

According to industry areas:

- **Vertical marketplaces**
  - Serve one industry
  - Automate supply chain
- **Horizontal marketplaces**
  - Span multiple industries
  - Help to mediate common services and goods

According to facilitator:

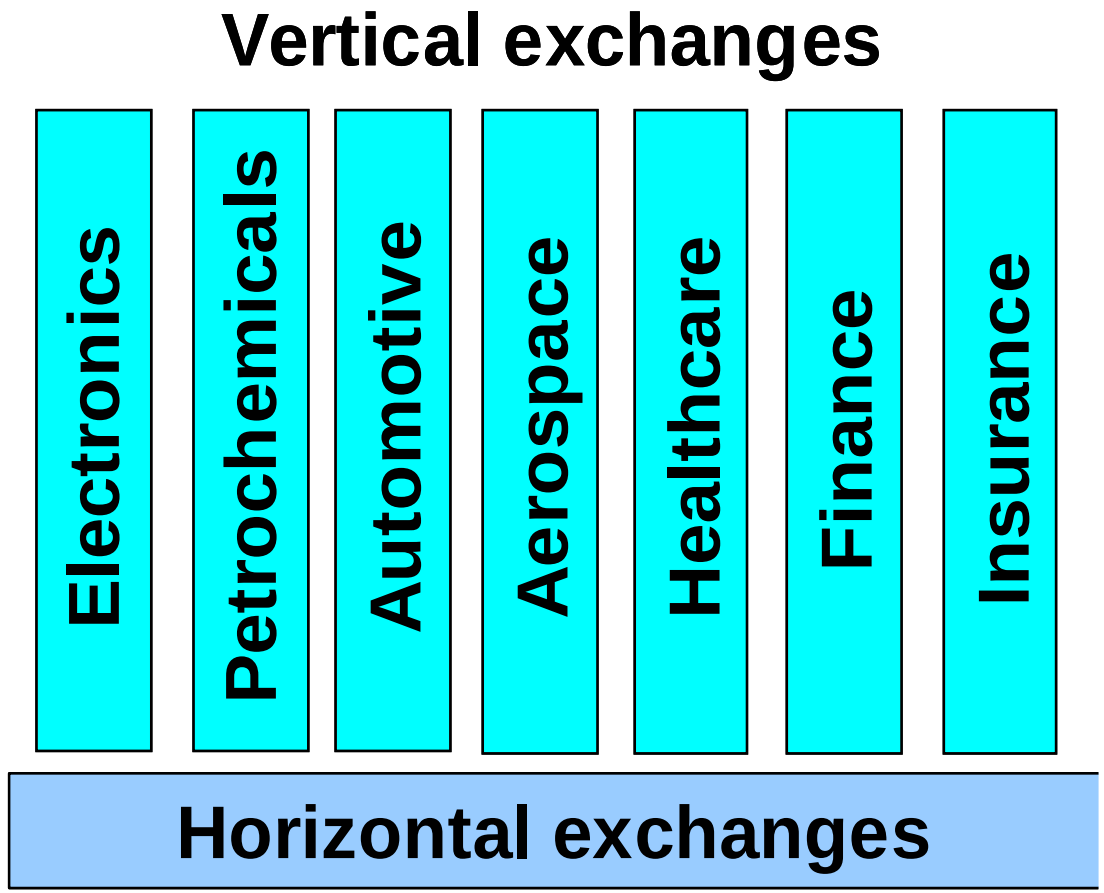
- Buyer oriented marketplaces
- Supplier oriented marketplaces
- Intermediary oriented marketplaces

According to number of participants

- Pyramid model
- Butterfly model

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Classification of E-marketplaces



Vertical: within an industry

Horizontal: participants from several different industries

Example of vertical marketplaces

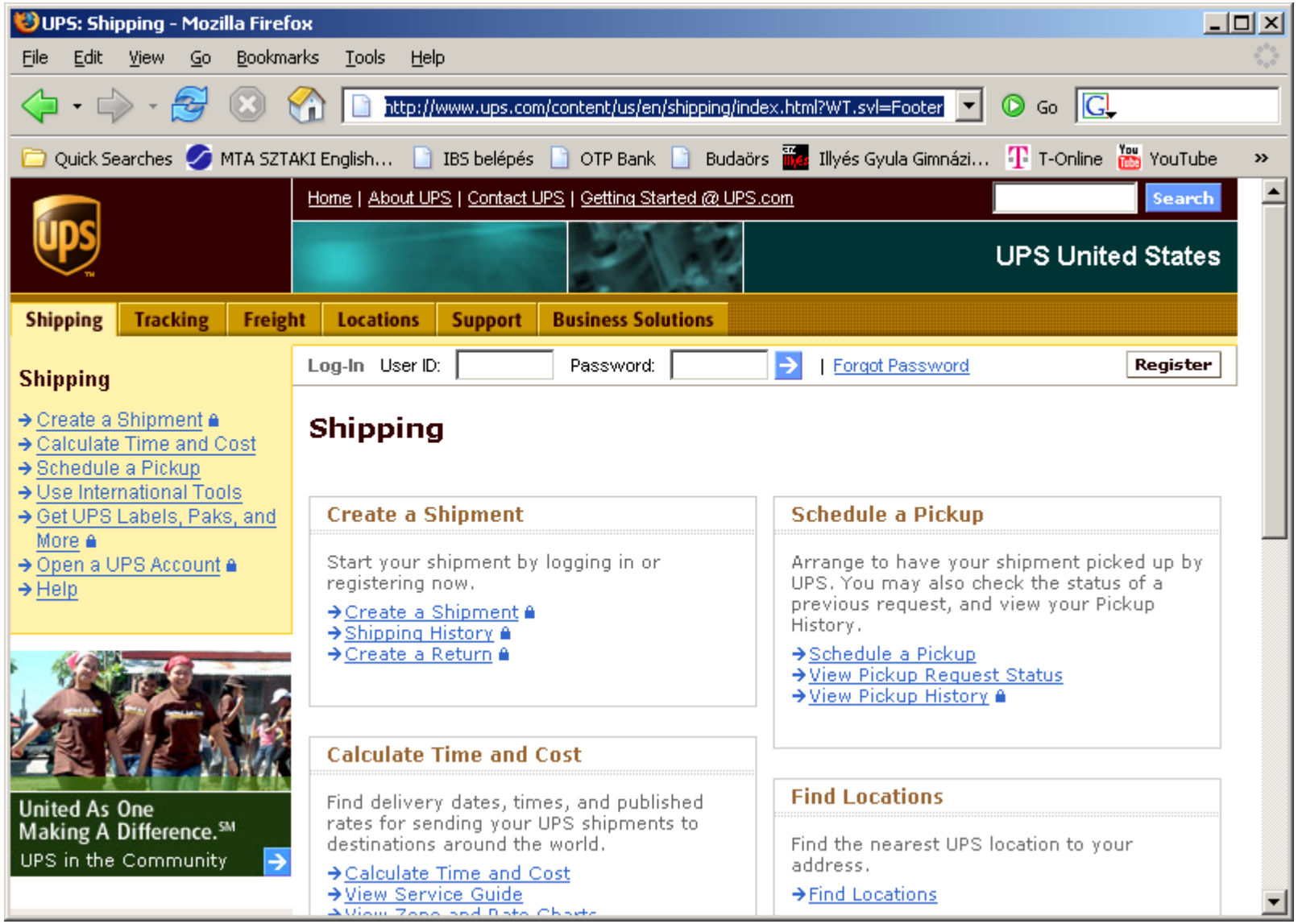
http://www.chemicalmarketplace.com/



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Example of horizontal marketplaces

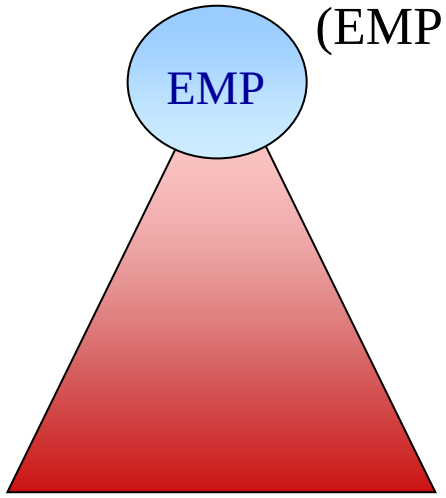
http://www.ups.com/



E-Marketplace Models

Pyramid model

Few buyers



(EMP: **e-m**arket**p**lace)

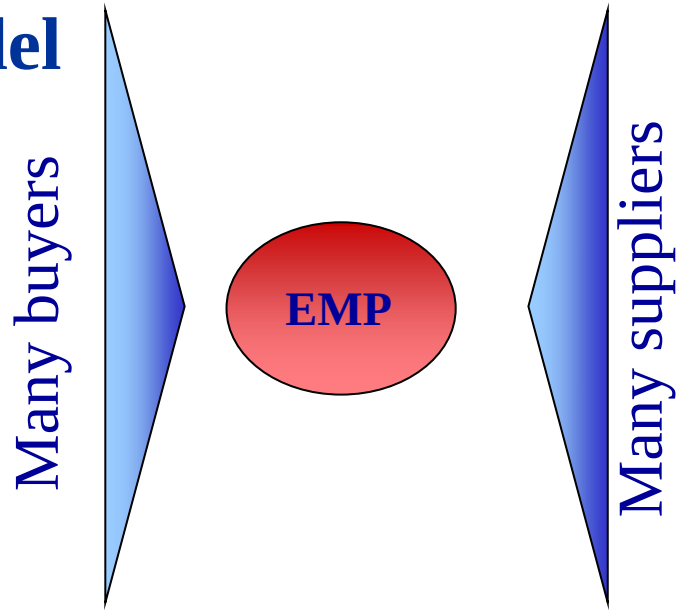
Many suppliers

- Large number of suppliers - few buyers (or the other way round)
- E.g.: automotive, aerospace, defence, metals industries
- Sites are an extension of supply chain for major enterprises to reduce transaction costs.
- Limited opportunity for e-marketplace to generate significant revenue

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E-Marketplace Models

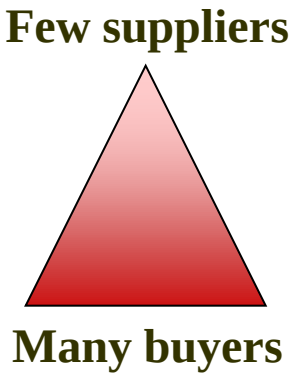
Butterfly model

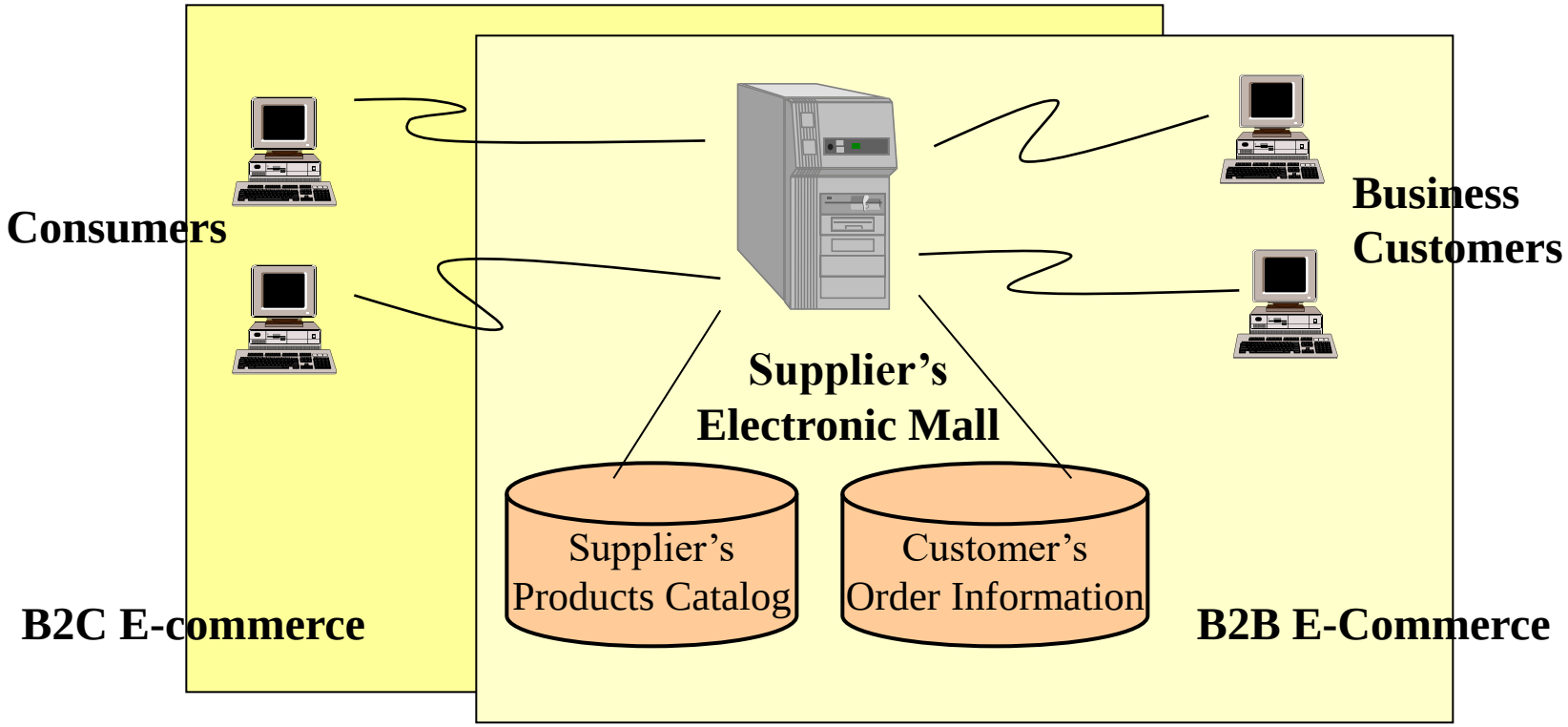


- Large number of buyers and suppliers
- E.g.: computing and electronics, utilities, food and agriculture industries
- dynamically evolving marketplaces, perfect competition, opportunity to generate high profit

Supplier-Oriented Marketplace

- Pyramid model
- A supplier or a consortium runs the site to sell via catalogues or forward auctions.
- Individual consumers and business buyers may use the same supplier-provided market place (May pay different prices due to quantity discount)





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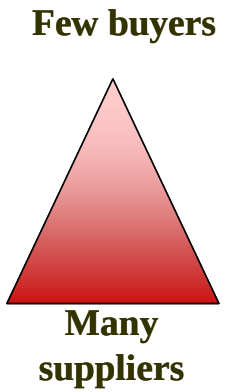
Example of supplier-oriented B2B marketplaces

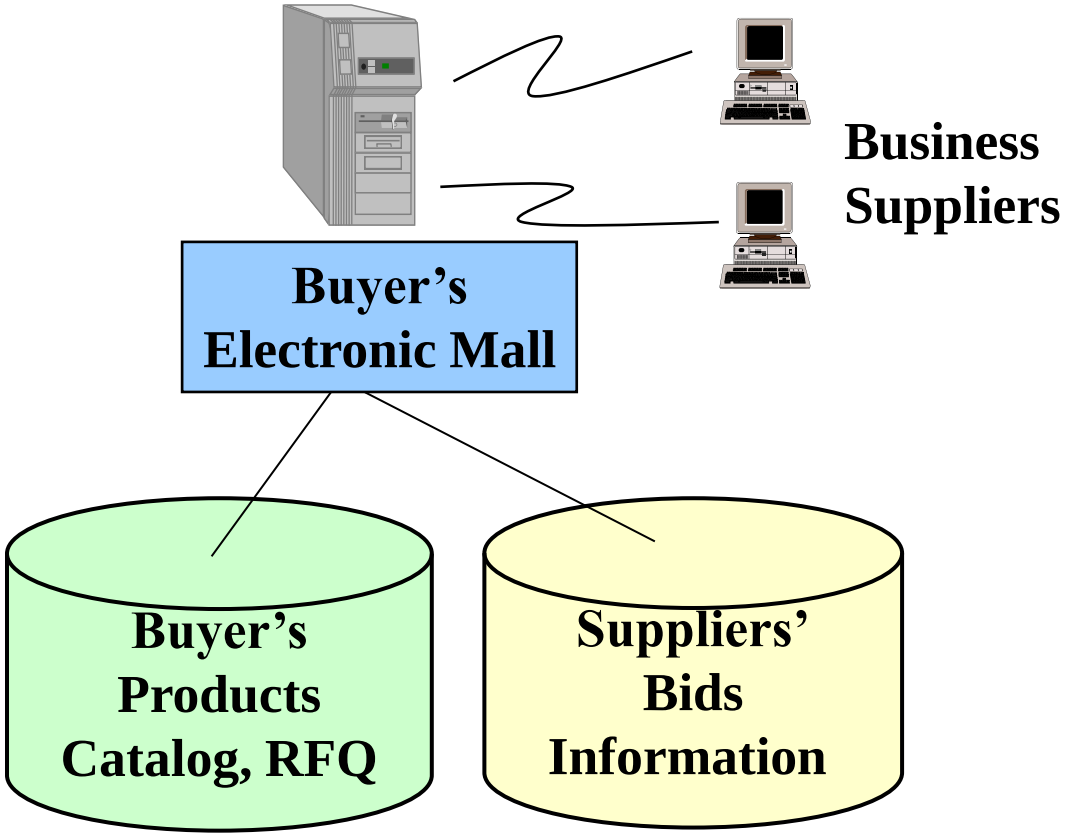
http://www.unitedaluminum.com/



Buyer-Oriented Marketplace

- Pyramid model
- Buyer opens a market on its own server
- Invites potential suppliers to bid on RFQs.
- Offers opportunity to committed suppliers
- Only very big buyers can afford

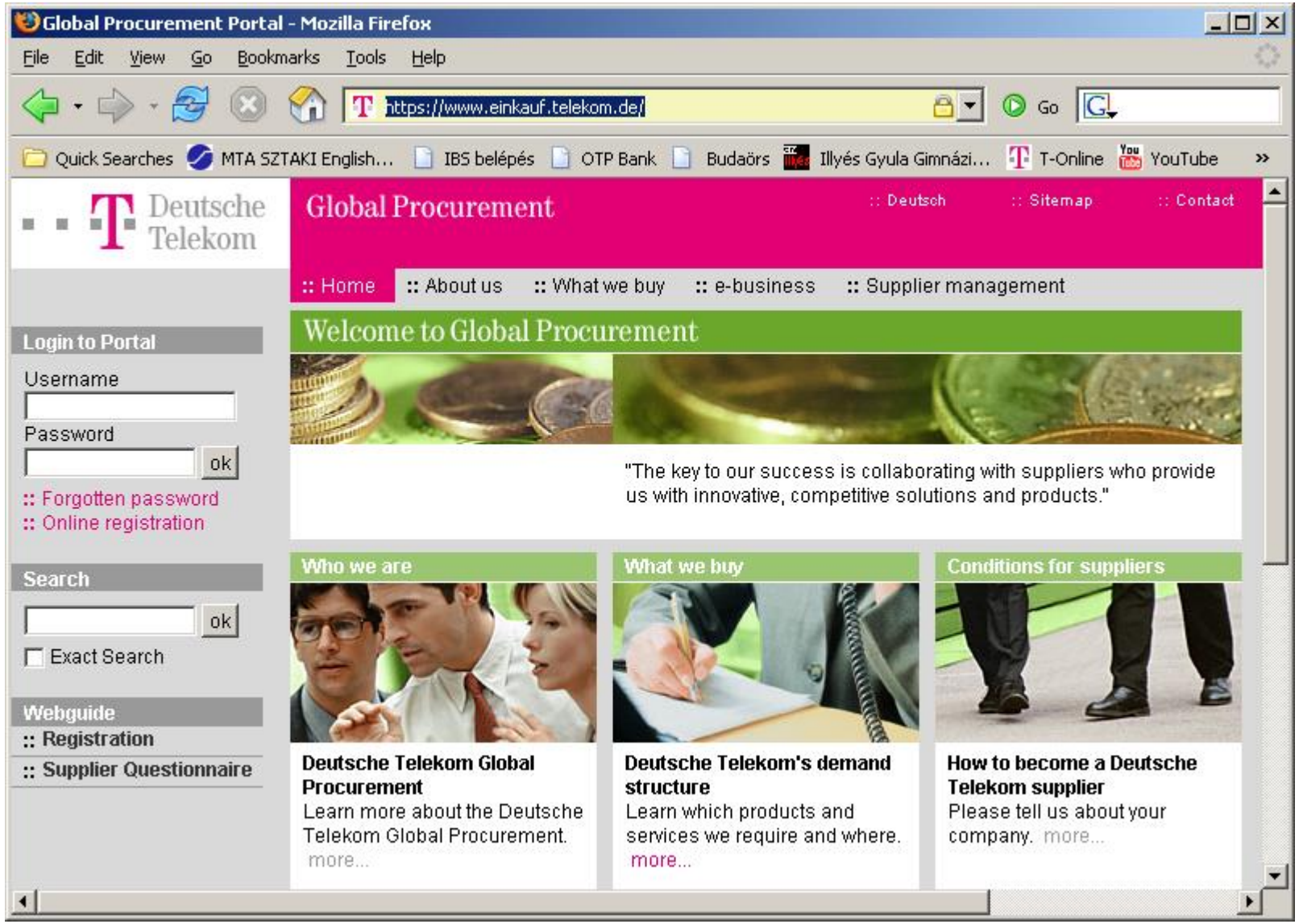




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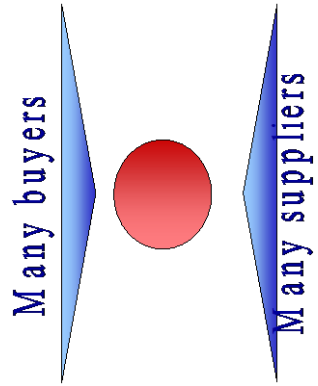
Example of buyer-oriented B2B marketplaces

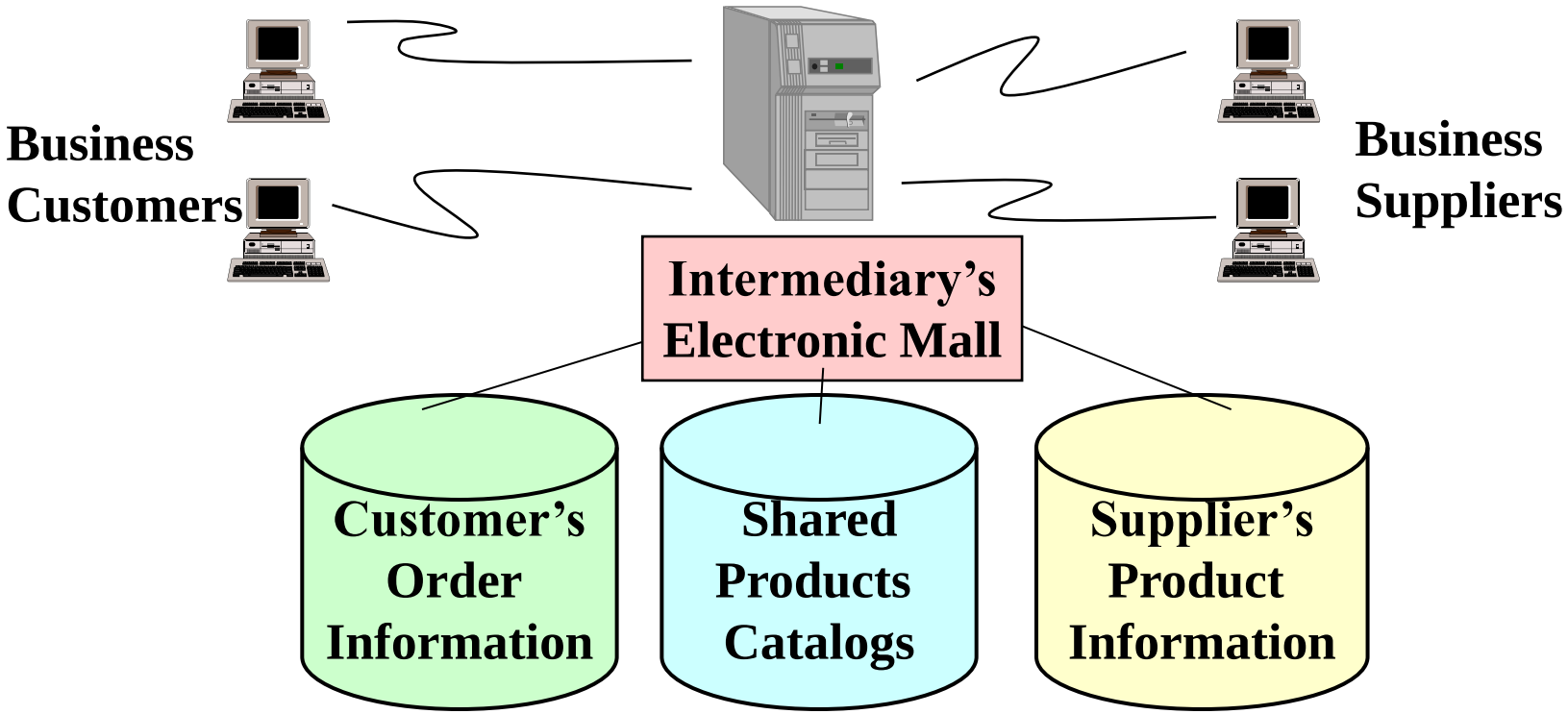
https://www.einkauf.telekom.de/



Intermediary-Oriented Marketplace

- Butterfly model
- Electronic intermediary company brings buyers and suppliers to one place
- Facilitates catalogues or auctions
- The corporate information systems need tight coupling with the intermediary's information system





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Example of intermediary marketplaces

http://b2bfoodmarket.com/

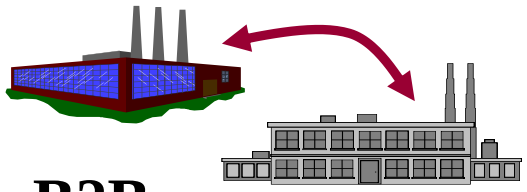


B2B versus B2C



B2C

- Business relationships are
  - simple
  - usually ad hoc
- No formal contracts
- Low switching costs for customers
- Customers look for attractions
- Low volume of transactions
- Open for everybody
- Trust is a problem



B2B

- Business relationships are
  - complex
  - often long term
- Formal contracts
- Need for strategic partnerships
- High requirements for integration
- Large transaction and payment volume
- Registration is a special issue
- Less problems with trust because of known partners

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Any questions?



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